

“A Study on Factors Influencing the Customer Buying Behaviour in Automobile Market with Special Reference to Government Policies and Incentives for Electric Vehicles in India”

I. INTRODUCTION:

The transportation plays a very important role in the development of Indian regions. The Indian automobile industry is a significant contributor to India's GDP and manufacturing sector. The use of transportation increased due to the human population has increased in India. The current transportation system totally depends on internal combustion engine vehicles called Commercial Vehicles (CVs). It is a cause of emissions of many harmful greenhouse gases such as carbon, which are associated with climate change and environmental pollution. The current environmental condition of the whole world is on red alert due to the continued emission of Greenhouse Gases (GHGs) and Global warming (GW). Under GHGs, CO₂ is under tremendous pressure to reduce carbon footprints. The temperature of a planet is continuously increasing because of GW. Fossil fuels include oil, coal, and natural gas, and burning them causes what is known as the “Greenhouse Effect” in Earth’s atmosphere.

The government of India makes recommendations in the year of 2011 to promote electric mobility and manufacturing of Electric Vehicles (EVs). EVs are the best example that helps reduce the emission of GHGs and GW emissions. The Indian government has come up with ambitious plans of introducing the EVs to Indian market and keep in pace with the development of EVs globally. The National Electric Mobility Mission Plan 2020 has come with a detailed report on EVs which the objective of promoting hybrid and electric vehicles. EVs can be turned into a major contributor to the successful implementation of the smart grid in the automobile industry. EVs are gaining significant attention and popularity worldwide due to their potential to reduce greenhouse gas emissions, dependence on fossil fuels, and air pollution.

EVs have developed as a potential solution to the rising environmental concerns associated with conventional gasoline-powered vehicles. Nashik, a rapidly developing city in Maharashtra of India, has a significant potential market for EVs, and understanding Factors Influencing the Customer Buying Behaviour of Automobile Market with Special Reference to Government Policies and Incentives for Electric Vehicles in India

Indian Government Policies & Incentives on EVs

The Government of India has implemented a comprehensive policy framework to promote the adoption of electric vehicles (EVs) in the country. The policy offers different types of financial incentives to make electric vehicles more affordable to all EV Buyers. The government aims to create a conducive environment for EV adoption, reduce reliance on imports, and foster a sustainable and green transportation system for the future.

Maharashtra Government Policies & Incentives on EVs

Maharashtra was one of the first states in India to design and notify an EV policy. Maharashtra's EV policy was released in February 2018. As per the update that, the Maharashtra Government updated & introduced the Electric Vehicle Subsidy Policy in July 2021. Due to this, the main objective is to promote the use of electric vehicles in Maharashtra state. Maharashtra Government offered incentives to purchase, road tax exemption, Charging Infrastructure, Interest Subsidy, and support to manufacturing, R&D.

II. RATIONALE AND SIGNIFICANCE OF THE STUDY

While there is ample research on customer buying behavior in the automobile market, there is a lack of comprehensive studies specifically focused on the Indian market with its unique characteristics. India's Central & State Government has their own set of policies and incentives for promoting EV adoption, and understanding how these factors influence customer buying behavior is essential for policy formulation and the Automobile industry's success and growth. The study can assess the impact of Government Policies and incentives on customer buying behavior. However, consumers' adoption of electric vehicles depends on several factors, including their attitudes toward the environment and purchasing behavior. By examining the effectiveness of these policies in influencing consumer decisions, the study can provide valuable insights into the strengths and weaknesses of current policies, identify gaps, and suggest improvements.

The significance of the study lies in its potential to provide insights into the factors that influence customer buying behavior in the automobile market with special reference to government policies and incentives for electric vehicles. This information can help the automobile industry and policymakers to develop effective strategies to promote the

adoption of electric vehicles in the automobile market of Nashik district. Additionally, the study can provide valuable insights for automobile companies looking to enter the electric vehicle market in Nashik district. By understanding the factors that influence customer behavior, these companies can design and market their products to meet the needs and preferences of consumers in the region.

Overall, the study has significant implications for both the environment as well as the economy of the automobile industry. By promoting the adoption of electric vehicles, the study can help to reduce carbon emissions, improve air quality, and mitigate the impacts of climate change. This study can create new economic opportunities for businesses in the electric vehicle industry in Nashik district.

III. RESEARCH PROBLEM STATEMENT:

The Indian government has implemented a comprehensive framework of various policies and incentives to promote the adoption of electric vehicles (EVs) in the automobile market of the country. However, the effectiveness of these policies in influencing customer buying behavior is still unclear. So, the aim of this study is to investigate the factors that influence customer buying behavior in the automobile market in India, with a specific focus on government policies and incentives for EVs.

This study will address the following research questions:

- What are the key factors that influence customer buying behavior in the automobile market in India?
- How do the Indian government policies and incentives for EVs impact on customer buying behavior?

By exploring these research questions, this research study aims to provide insights into the impact of government policies and incentives on customer buying behavior in the Indian automobile market.

IV. OBJECTIVES:

The main objectives of this study are as follows:

1. To identify the Marketing factors influencing customer buying behavior in the automobile market specifically focusing on electric vehicles.
2. To examine the impact of government policies and incentives on customer buying behavior towards electric vehicles.
3. To explore the role of demographic factors in customer buying behavior towards electric vehicles.

4. To identify the Infrastructural barriers faced by customer in adopting electric vehicles.
5. To analyze the challenges faced by customer in the adoption of electric vehicles.
6. To suggest remedial measures to promote the adoption of electric vehicles for policymakers, the automobile industry, and other stakeholders based on the findings of the study.

V. HYPOTHESIS:

Hypothesis testing is a statistical approach used to test a hypothesis or assumption about a population based on a sample of data. In the context of A Study on Factors Influencing the Customer Buying Behaviour in Automobile Market with Special Reference to Government Policies and Incentives for Electric Vehicles in India, a possible hypothesis to test could be:

H1: There is a significant impact of Marketing Factors on the customer buying behavior about Electric Vehicles.

H2: Government policies and incentives for Electric Vehicles in India have a significant influence on customer buying behavior in the automobile market.

H3: There is a significant relationship between the demographic factors of the customers and their buying behavior about Electric Vehicles.

H4: Infrastructural barriers significantly influence customer buying behavior for Electric Vehicles.

VI. RESEARCH METHODOLOGY:

This study will use a mixed-methods research approach, including both qualitative and quantitative approaches. The study will begin with a literature review of the applicable theories and empirical studies related to the factors that influence customer attitudes toward the adoption of Electric vehicles. This will inform the development of the research questionnaire and interview questions. The research will be conducted with a representative sample of customers in the Nashik district. The primary data will depend on the Environmentally Responsive Buying Behaviour of Customers towards the

Adoption of Electric Vehicles, including their perceived benefits, concerns, and preferences. The survey will also collect demographic information, such as age, gender, income, education, and occupation. The interviews will be conducted among a convenient purposive sample of electric vehicle owners and potential electric vehicle buyers. The interviews will be conducted to gain a deeper understanding of the Buying Behaviour of Customers toward the Adoption of Electric Vehicles and their factors. The interviews will be conducted through in-person and telephonic immense methods and it is based on the participants' preferences.

a. Research Design:

Exploratory and Descriptive Research Design

Exploratory research Design is an examination of a subject in an attempt to gain further insights in the early phase of the research. A researcher would like to explore various insights through exploratory research design.

Descriptive Research Design describes what exists and may help to uncover new facts and meaning so the researcher collects information with the help of questionnaires and using participative observation through Individual interactions.

Qualitative and Quantitative Research Approaches:

A qualitative research approach is used to explore and understand the factors influencing customers buying behavior toward electric vehicles. This will help uncover customers' attitudes, perceptions, and experiences related to electric vehicle adoption.

A quantitative research approach is used to measure and quantify the extent of customers buying behavior toward electric vehicles. This will provide a broader understanding of the overall adoption patterns and preferences.

b. Sources of data collection:

The data will be collected by both primary and secondary means:

- i. Primary Data** will be collected by questionnaires and using participative observation through Individual interactions.

- ii. **Secondary Data** will be collected through Previous Research thesis, Government records, Annual Reports of the companies, books, articles, conference publishing, newspapers, internet browsing, Reports, and Publications, etc.

c. Sample Design:

- **Sample Population:** The sample population will be electric vehicle owners and potential electric vehicle buyers.
- **Sample Area:** The sample area will be Nashik District.
- **Sample Size:** Considering the Reliability level 95% or significance level 0.05 for the large population. The sample size has been calculated using the following Cochran's sample size formulae:

$$n = \frac{Z^2 pq}{E^2}$$

where: n = number of items in a sample,

Z^2 = square of confidence level in standard error units

p = estimated proportion of success

q = 1-p or estimated proportion of failures

E^2 = square of maximum allowance for error between true proportion and the sample proportion

Therefore:

$$n = \frac{(1.96)^2 \times 0.7 \times 0.3}{(0.05)^2}$$

$$= 322.69 \text{ units}$$

- **Sample Technique:** Will be Multistage Sampling
 - **Convenient Purposive Sampling** is a non-probability sampling technique that allows researchers to select participants based on their accessibility and availability.

- **Random Sampling** is a sampling technique that involves selecting samples from a population in a random manner with an equal chance of being included in the research.

d. Statistical Tools to Be Used:

- **Statistical Tools & Software:** SPSS, AMOS, Ms-Excel, and Suitable statistical tools will be used for the purpose of data analysis and testing of the hypotheses.
- The collected primary data will be presented in the form of charts, tables, diagrams, and as per the requirement of testing the Hypothesis by using relevant statistical techniques and it will be presented in the form of T-Test/ correlation analysis/ one-way ANOVA analysis/ Simple Comparison/ Percentage, etc. whichever be suitable for the collected data & need of the analysis.

e. Scope of The Study:

- The study will investigate a wide range of factors that impact customer buying behavior of EVs in the automobile market.
- The study will examine the impact of government policies and incentives on customer buying behavior of EVs.
- The study will primarily focus on the Indian market and the specific government policies and incentives related to EVs implemented in India.

f. Limitations of The Study:

- The study will be limited to the area of Nashik District.
- Time is always a limiting factor for research.
- The information collected for research will be totally dependent on the judgments and capability of the respondent.

VII. LITERATURE REVIEW:

H. Nurgul Durmus Senyapar and others (2023):

This study aims to examine the various policies and incentives for EV purchase in different countries, evaluate consumer concerns prior to the adoption of smart devices

(SDs), and assess consumer intentions towards EV adoption using theoretical models such as planned behavior theory, reasoned action theory, and technology acceptance model. Consumers' intentions and exploring strategies to expand the EV market in developing countries, such as Turkey, remain important research areas of investigation. Additionally, evaluating consumer concerns prior to adopting SDs helps to identify challenges that need to be addressed to increase consumer acceptance of EVs. By using some theoretical models, the study assesses consumer intentions towards EV adoption and provides insights into the psychological and behavioral factors that influence consumer decision-making. The findings of this study shed light on important factors that can help increase the market share of EVs, which play a crucial role in reducing carbon footprint.

Ishika Ranjan and others (2022):

This research focuses on analyzing consumer demand for EVs by investigating the impact of consumer innovation and concerns about the functional capabilities of EVs and their preferences. To accomplish this, a conceptual framework is developed and implemented, incorporating measures of innovativeness at the adoption level. These measures are based on an evaluation of technology ownership and social factors. In India, EVs are required because fuel prices are continuously hikes so the automobile industry has introduced EVs as a potential solution for both the industry and the environment which follows all government rules and regulations. Despite the implementation of government policies supporting EVs, the current market penetration of these vehicles remains low. Therefore, this research aims to examine consumer buying behavior and perceptions towards EVs in India. The target respondents for data collection will include two-wheeler original equipment manufacturer (OEM) dealers, students, and working employees. The end consumers targeted for the study will be from tier 1 and tier 2 cities. This research is to gain insights into consumer behavior and preferences regarding EVs in the Indian market. The findings will contribute to a better understanding of the factors influencing consumer adoption of EVs and help identify potential strategies to promote their uptake.

Mr. Channabasav Hiremath and others (2022):

This research is focused on consumer buying behaviour towards EVs and the main objective is to examine the shift in the automotive sector from conventional vehicles to electric vehicles (EVs) based on consumer preferences with the factors such as the rapid increase in fuel prices and concerns about environmental pollution. Automakers are now

planning to increase the annual production of EVs to meet the growing demand as the requirement of customers. In this study, findings were obtained through a survey conducted in Bagalkot City, with 100 respondents participating. Primary data is collected through structured questionnaires administered to customers who express an interest in purchasing EVs in the future. Consumer purchase decisions about EVs are influenced by various internal and external factors. The study highlights that most of the respondents are aware of the two EV models i.e. Hero and Ather models. Furthermore, the study reveals that many of the customers fall within the age category of 36-45 years. The findings will provide a better understanding of the factors influencing customers' decisions to adopt EVs and help identify key drivers for EV market growth in the region.

S. Goel and others (2021):

This research has been conducted on the attributes and characteristics of EVs and their charging infrastructure, the production and network modeling of electric vehicles continue to evolve and face constraints. This study provides an overview of studies focusing on the penetration rate of EVs, HEVs, PHEVs, and BEVs in the market. It also discusses the various optimization techniques and approaches employed in these studies. The research mainly addresses the unique challenges and barriers related to insufficient charging facilities in a developing country like India. Additionally, the paper introduces the concept of Vehicle-to-Grid (V2G), which enables electric vehicles to serve as an additional power source when renewable energy sources are unavailable. Overall, the research highlights the special characteristics of electric vehicles in mobility planning. It emphasizes the need to address barriers and promote the widespread adoption and efficient operation of electric vehicles. Automobile Industry offers a sustainable alternative to traditional vehicles by reducing dependency on fossil fuels and promoting the deployment of renewable energy sources on a large scale.

Henry Lee and others (2018):

This paper aims to analyze the further advances that are necessary to facilitate the significant penetration of EVs into the passenger vehicle fleet. To achieve widespread adoption of EVs, several key areas need to be addressed and improved upon. There are two important crucial factors addressed in this study i.e., reducing the upfront cost of EVs & the expansion and enhancement of charging infrastructure. Reducing the upfront cost of EVs involves continuing to decrease the cost of battery production through

technological innovations and economies of scale. The availability of convenient and fast charging stations is essential to alleviate concerns about range anxiety and provide EV owners with the necessary infrastructure for long-distance travel. The paper emphasizes that to achieve significant penetration of EVs in the passenger vehicle fleet, a multidimensional approach is required. It necessitates not only ongoing advancements in battery technology but also comprehensive strategies to address cost barriers, expand charging infrastructure, and educate and incentivize consumers to adopt EVs.

Prof. Rajesh and others (2022):

This study focuses on the factors influencing the adoption of electric vehicles in India, including financial incentives, charging infrastructure, social reinforcement, environmental concern, and price. The study collected data from 150 randomly selected respondents across India. Confirmatory Factor Analysis (CFA) was employed to analyze the data and examine the relationship between the independent variables and the dependent variable, which is electric vehicle adoption. The increasing presence of major car manufacturing giants in the electric vehicle (EV) space indicates EVs' growing importance and popularity. As a result, there is a need for more sustainable and environmentally friendly modes of transportation. EVs have emerged as a potential solution, but the adoption rate in India has been relatively low. This research suggests that the affordability and competitive pricing of electric vehicles play a crucial role in encouraging their adoption by consumers. Understanding the factors, this study provides insights into the barriers and opportunities for increasing the adoption rate in India. The study emphasizes the need for competitive pricing and financial incentives to accelerate the transition to EVs and create a more environmentally friendly transportation system.

Dutta B. and others (2021):

The purpose of the current study is to explore barriers influencing consumers' intention to adopt sustainable electric vehicles (EV) based on the modified theory of planned behavior (TPB) model. The findings of the current study explored how attitude (ATT), subjective norm (SN), and perceived behavioral control (PBC) significantly influenced users' sustainable consumption intentions. The finding also explored ways in which environmental concern significantly influenced ATT, SN, PBC, and the sustainable consumption intention of the users. Not only measurements of vehicle performance, namely safety, reliability, and range, but other factors, such as purchasing price, charging

facility, and maintenance and battery cost also influenced consumers' sustainable consumption intentions. The findings of the study could improve consumers' sustainable consumption intentions to prevent further air pollution and reduce CO2 emissions from the transportation sector.

VIII. EXPECTED OUTCOMES:

The study may contribute to increasing the awareness and knowledge among customers about the Government benefits of electric vehicles (EVs) compared to conventional gasoline-powered vehicles. It may help in educating customers about the positive impacts of EV adoption on reducing greenhouse gas emissions and air pollution. The proposed study on Factors Influencing Customer Buying Behaviour in Automobile Market with Special Reference to Government Policies and Incentives for Electric Vehicles in India holds significant potential for understanding the factors influencing EV adoption and promoting sustainable transportation practices. The study can assess the level of environmental concern among customers and how it relates to their buying behavior toward EVs.

Overall, this research proposal holds promise for contributing to the knowledge base on customer buying behavior towards EV adoption with reference to Government Policies in the Nashik District.

IX. SCHEME OF CHAPTERS:

Sr. No	Particulars
1	Chapter I: Introduction: 1.1.An overview of the Indian Automobile Industry. 1.2.Evolution of EVs in India. 1.3.Government Initiatives for EV & Indian Auto Customers.
2	Chapter II: Literature Review: 2.1 Evolutions Stages in Auto Sectors. 2.2 Consumer Behaviour in Auto Industry. 2.3 Study of Relevant Literature.
3	Chapter III: Theoretical Background: 3.1 Customer Behaviour. 3.2 Government Policies (Central & State) to Promote EVs.

4	Chapter IV: Research Methodology: 4.1 Research Design 4.2 Types of Data 4.3 Sampling Method
5	Chapter V: Data Analysis & Interpretation: 5.1 Reliability & Validity of Data. 5.2 Data Presentation 5.3 Data Analysis 5.4 Hypothesis Testing
6	Chapter VI: Findings, Conclusion, and Suggestions of the Study
	ANNEXURE - Questionnaire, - Bibliography - Government Reports

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Mr. Ankush Pingale
Research Student